



Hands-On Activity: Calculations with SQL



Activity Overview

Previously, you learned about basic calculations with SQL. Just like spreadsheets, you can use the four basic arithmetic operations to perform addition, subtraction, multiplication, and division. Now, you'll practice writing basic calculations using single and multiple operators in Bigquery. This will enable you to combine multiple arithmetic operations in a single query, allowing you to quickly and efficiently discover significant patterns in your data.

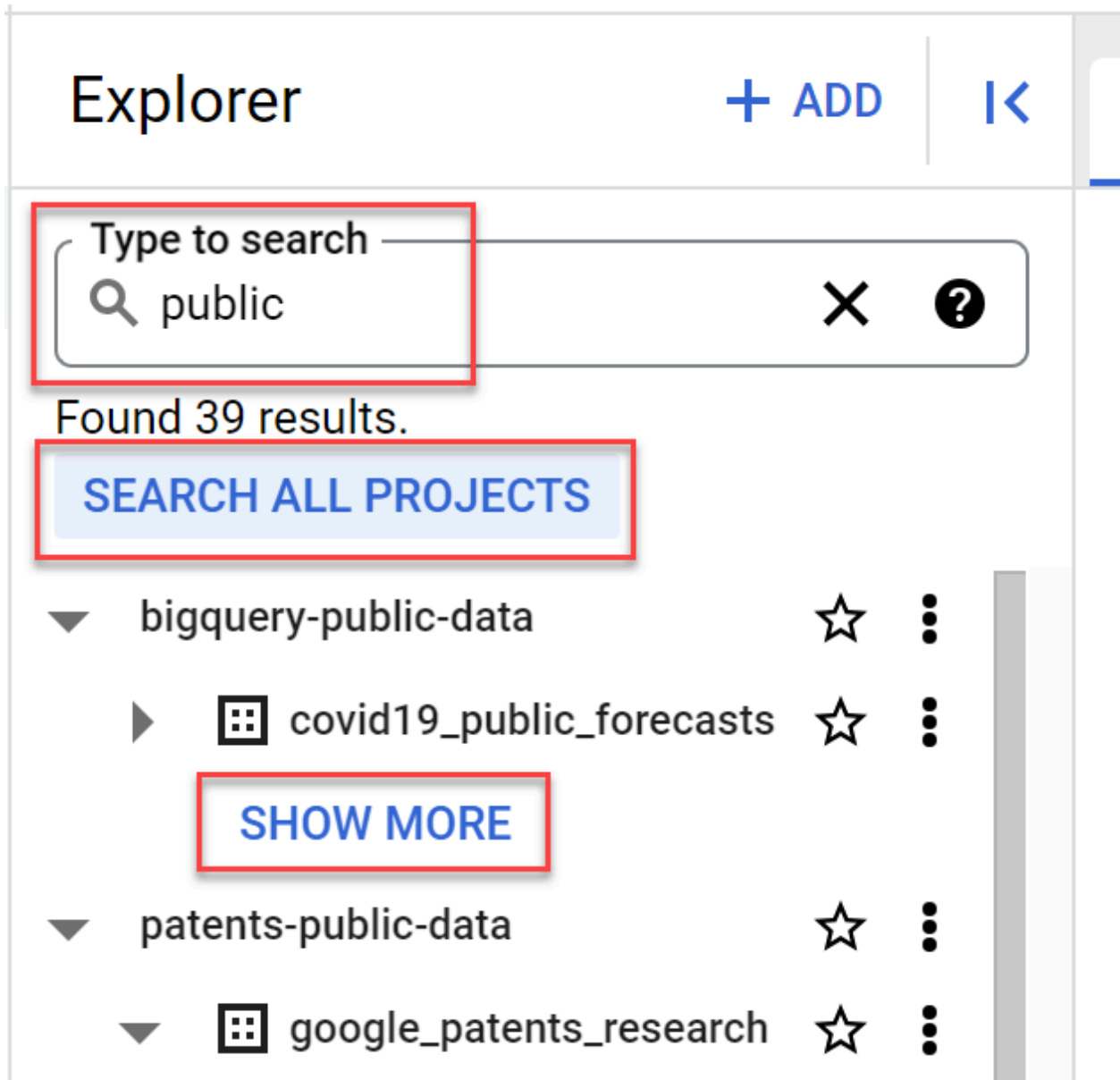
Follow the instructions to complete each step of the activity. Then answer the question at the end of the activity before going to the next course item.

For this activity, you'll analyze subway ridership data to help improve the quality of the city's public transportation. To do this, you'll use data that describes the average weekly subway ridership in New York City from 2013-2018.

1. Navigate to your [BigQuery console](#).

- **Note:** BigQuery Sandbox frequently updates its user interface. The latest changes may not be reflected in the screenshots presented in this activity, but the principles remain the same. Adapting to changes in software updates is an essential skill for data analysts, and it's helpful for you to practice troubleshooting. You can also reach out to your community of learners on the discussion forum for help.

1. Locate the public data by **typing** the word public in the **search bar**. Select **SEARCH ALL PROJECTS** and **SHOW MORE** to access all of the public datasets.



2. Select the **star icon** next to bigquery-public-data. This will pin it to the Explorer menu.

Explorer

+ ADD



Type to search

public



Found 39 results.

[SEARCH ALL PROJECTS](#)

▼

bigquery-public-data

★

⋮

▶

 covid19_public_forecasts

☆

⋮

[SHOW MORE](#)

▼

patents-public-data

☆

⋮

▼

 google_patents_research

☆

⋮

 publications

☆

⋮

3. Select the **dropdown arrow** next to the bigquery-public-data to expand all the available datasets.

4. Navigate down the list until you find the **new_york_subway** dataset.

Note: You may have to select the **SHOW MORE** button twice to reach the correct dataset. Now, select the **dropdown arrow** next to new_york_subway and select **subway_ridership_2013_present**.

Explorer

+ ADD



Type to search

public



Found 3 results.

SEARCH ALL PROJECTS

- ▶  ncaa_basketball  
- ▶  nces_ipeds  
- ▶  new_york  
- ▶  new_york_311  
- ▶  new_york_citibike  
- ▶  new_york_mv_collisions  
- ▼  new_york_subway  
 -  geo_nyc_borough_bo...  
 -  routes  
 -  station_complexes  
 -  station_entrances  

Note: This dataset may not be accessible through other search methods. Follow the previous steps to locate the correct data for this activity.

Note also that any of the public databases on BigQuery are living records and, as such, are periodically updated with new data. Throughout this course gram), if your results differ from those you encounter in videos or screenshots, there's a good chance it is due to a data refresh.

First, identify data exploring the change in weekly ridership from 2013 to 2014. Use SQL to subtract the number of riders in 2013 from the number of riders in 2014.

1. Select the **dropdown arrow next to QUERY** and select **In split tab**. You will then be able to see the Query editor and the datasets. If you prefer to only see the Query editor, you could select **In new tab**.

subway_ridership_2013_p...ent

subway_riders...

QUERY

In new tab

In split tab

Filter Enter property name or value

☐	Field name	Type	Mode
☐	station_name	STRING	NULLABLE
☐	routes	STRING	NULLABLE
☐	ridership_2013	INTEGER	NULLABLE
☐	ridership_2014	INTEGER	NULLABLE
☐	ridership_2015	INTEGER	NULLABLE
☐	ridership_2016	INTEGER	NULLABLE

2. In the Query editor, add the names of the columns you want to use in your calculations in your **SELECT** statement. In this case, select the following columns:

- station_name
- ridership_2013
- ridership_2014

Be sure to add a comma after each column name. 2. Add the calculation to the query by adding the names of the two columns with a minus sign between them.

- ridership_2014 - ridership_2013

3. List the result in a new column. To do this, enter **AS** followed by the new column's name. Name it **change_2014_raw** because it represents the change in ridership from 2013 to 2014 in raw numbers.

4. End your query with the **FROM** command and the name of the dataset and subset you're pulling data from. After **FROM**, select **Enter** or **Return** and enter **bigquery-public-data.new_york_subway.subway_ridership_2013_present**.

SELECT

station_name,

```
ridership_2013,
```

```
ridership_2014,
```

```
ridership_2014-ridership_2013 AS change_2014_raw
```

```
FROM
```

```
bigquery-public-data.new_york_subway.subway_ridership_2013_present
```

5. Select **RUN** to get the results.

The results show the change in ridership from 2013 to 2014 at different stations. For example, the Atlantic Av - Barclays Ctr station gained an average of 1,774 riders per week. The 4 Av station lost 321 riders.

By including a basic calculation in your query, you can quickly gain important knowledge about your data. In this example, you now have insights into the change in ridership for each subway station in any given year.

Query results					SAVE RESULTS EXPLORE DATA	
Query complete (0.0 sec elapsed, cached)						
Job information					Results JSON Execution details	
Row	station_name	ridership_2013	ridership_2014	change_2014_raw		
1	Atlantic Av - Barclays Ctr	39871	41645	1774		
2	4 Av	13156	12835	-321		
3	14 St / 6 Av	49316	49990	674		
4	Jamaica - Van Wyck	4992	5121	129		
5	Crown Hts - Utica Av	27780	28287	507		

Query results showing Row 1, Atlantic Av - Barclays Ctr, ridership 2013 is 39871, ridership 2014 is 41645, change 2014 raw is 1774; Row 2, 4 Av, ridership 2013 is 13156, ridership 2014 is 12835, change 2014 raw is -321; Row 3, 14 St/6Av, ridership 2013 is 49316, ridership 2014 is 49990, change 2014 raw is 674; Row 4, Jamaica - Van Wyck, ridership 2013 is 4992, ridership 2014 is 5121, change 2014 raw is 129; Row 5, Crown Hts - Utica Av, ridership 2013 is 27780, ridership 2014 is 28287, change 2014 raw is 507.

Next, find the average weekly ridership for a longer period of time, such as the multi-year period from 2013-2016.

To do this, combine multiple arithmetic operations in a query. The average of a set of numbers is the sum of the numbers divided by the total number of values in the set. There are four values in your new set (ridership data for 2013, 2014, 2015, 2016). Use SQL to sum the numbers for each year and divide that sum by 4.

1. Select the **COMPOSE NEW QUERY** button to refresh the query editor.

2. Enter **SELECT** to select the columns you want to pull from the table. You're selecting several columns, so press **Enter** or **Return** after **SELECT** and add a comma after each column name. The columns selected for this query will be:

- enter station_name
- ridership_2013
- ridership_2014
- ridership_2015
- ridership_2016

3. Add the calculation to the query. Use parentheses to control the order of the operations if you use more than one arithmetic operator in a calculation. In this case, sum the years, then divide the sum by 4. Put parentheses around the sum of the four column names. Enter **(ridership_2013 + ridership_2014 + ridership_2015 + ridership_2016)**. Then enter a division operator **/** and the number **4**.

4. List the result in a new column by entering **AS** followed by the new column's name. Name the new **average** because it represents average weekly ridership for the period 2013-2016.

5. End your query with the **FROM** command and the name of the dataset and subset that you're pulling data from. After **FROM**, select **Enter** and type `bigquery-public-data.new_york_subway.subway_ridership_2013_present`.

SELECT

`station_name,`

`ridership_2013,`

`ridership_2014,`

`ridership_2015,`

`ridership_2016,`

`(ridership_2013+ ridership_2014+ ridership_2015 + ridership_2016) / 4 AS Average`

FROM

`bigquery-public-data.new_york_subway.subway_ridership_2013_present`

6. Select **RUN** to get the results. The results clearly show the trend in ridership at each station from 2013 to 2016. For example, weekly ridership at the Atlantic Av - Barclays Ctr station increased every year since 2013. Further, for the years 2014, 2015, and 2016, weekly ridership at Atlantic Av - Barclays Ctr exceeded the overall average for the period 2013-2016 (listed in the average column).

Query results						
 SAVE RESULTS  EXPLORE DATA ▼						
Query complete (0.3 sec elapsed, 19.1 KB processed)						
Job information <u>Results</u> JSON Execution details						
Row	station_name	ridership_2013	ridership_2014	ridership_2015	ridership_2016	average
1	Atlantic Av - Barclays Ctr	39871	41645	42231	42711	41614.5
2	4 Av	13156	12835	13126	13116	13058.25
3	14 St / 6 Av	49316	49990	50335	50692	50083.25
4	Jamaica - Van Wyck	4992	5121	5092	5022	5056.75
5	Crown Hts - Utica Av	27780	28287	28146	29069	28320.5

This kind of data is useful for managing public transportation. It can help you determine which stations or routes to expand due to increased ridership. Using basic calculations in your query allows you to quickly discover significant patterns in your data.

1. Reflection

1 / 1 point

Now you've spent some time performing calculations using SQL. In the text box below, write 2-3 sentences (40-60 words) in response to each of the following questions:

- When working with datasets, when might you use basic SQL calculations?
- How can those basic SQL calculations help make sense of large quantities of data?

My Answer: Here I use the basic SQL calculation when I use change_raw_2014 as a subtraction and average as the average of 2013 to 2016 years' average ride. On average, I did addition and then division. These 2 types of basic SQL calculations I have done in this lesson.

Those basic SQL calculations help me a lot in the 59M datasets. I have done the subtraction and gotten insight into the change in the raw 2014 column very easily. And also when I use the average operation, then I also get the proper insight of 59M of data of a new column of average.

Feedback

An effective reflection would note that you can use basic calculations in your query to help you quickly discover significant patterns in your data. While this example used public transportation, queries can be used with all sorts of datasets to provide you with useful information and patterns.